



Carbon footprint

In person teaching

Olivier Jolliet, DTU-Sustain, Quantitative Sustainability Assessment

In person teaching session – format & learning objectives

- ▶ Plenum lecture (OJ): Understand why and how to make a carbon footprint (10')
- ▶ Group interpretation (All): Interpret the car example results (10')
- ▶ Plenum OJ: Introduce input data for the morning coffee exercise. (5')
shortly present the morning coffee worksheet
- ▶ Groups work :
Perform the carbon footprint of a morning cup of coffee (40')
- ▶ Plenum:
Exercise overview (15')
- ▶ Link to your case study (TA): introducing the case study specific template or available material (10')



in Module 8, you will be able to apply the carbon footprint approach to your case study!



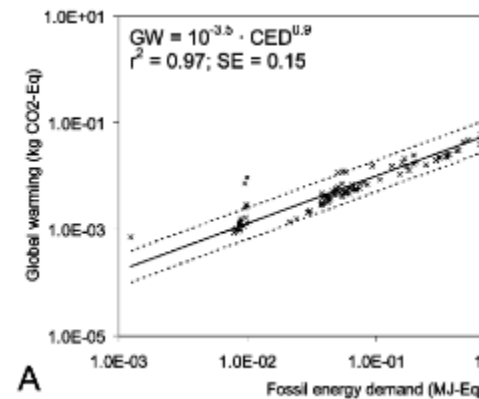
Determining the carbon footprint of human activity

Why and how

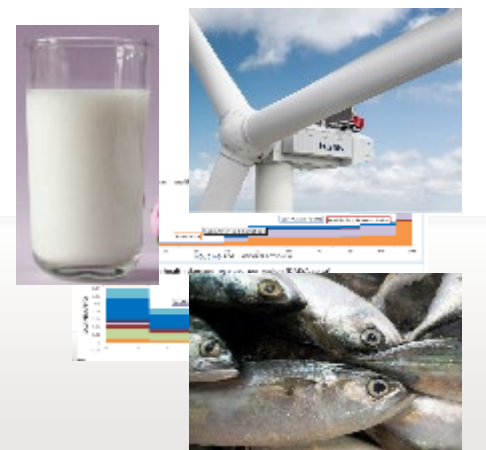
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Why performing a carbon footprint

- Because climate change has among the **highest environmental impact on humans and ecosystems**, especially in developing countries
- Because **many other environmental impacts are correlated to climate change**
→ co-benefits for many other types of impacts (except water and sometimes land use)
... and once you get the carbon footprint tool, you get impacts on humans & ecosystems
- Because **it is useful**: It helps identify hotspots in production process and compare **products & scenarios** per unit of consumption/function
- Because **we have a lot of data available to quantify the GHG emission** for more than 20,000 processes
- Because **it is relatively simple**: every engineers and everybody able to do multiplication and additions can do it
- Because **you will be able to apply it in your case study** in module 8 and then **use it in your professional life**
- Because **you will use the carbon footprint tool for the other modules and at the exam.**

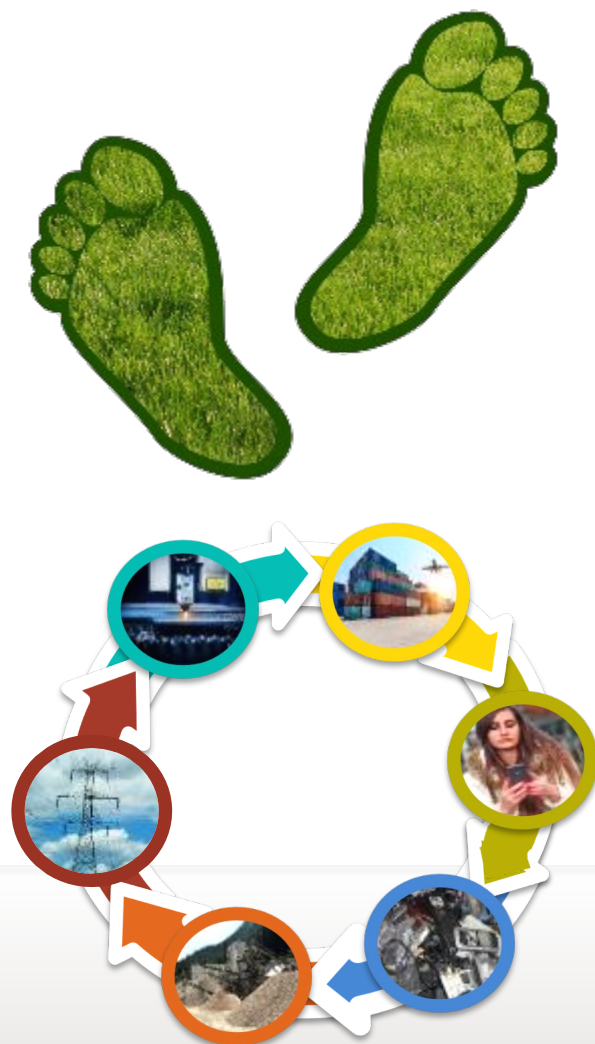


A screenshot of a software interface displaying a table of processes and their associated GHG emissions. The table has multiple columns, including process names and numerical values. The interface is labeled 'A' in the bottom left corner.



Carbon Footprint: at personal and product levels

A **carbon footprint** is the weighted sum of greenhouse gas (GHG) emissions caused by an individual, event, organization, service, or product, expressed as carbon dioxide equivalent. GHG emissions are weighted based on their global warming potentials (GWP100, as defined by the Intergovernmental Panel on Climate Change, AR6, Tables 7.15 and 7.SM.7)



Greenhouse Gas	GWP100
Carbon Dioxide (CO ₂)	1 kg _{CO2e} /kg _{CO2}
Methane biogenic (CH ₄)	27 kg _{CO2e} /kg _{CH4}
Methane fossil (CH ₄)	29.8 kg _{CO2e} /kg _{CH4}
Nitrous Oxide (N ₂ O)	273 kg _{CO2e} /kg _{N2O}
Hydrofluorocarbons (HFC-32)	771 kg _{CO2e} /kg _{HFC}
Perfluorocarbons (PFC-14)	7380 kg _{CO2e} /kg _{PFC}
Sulphur Hexafluoride (SF ₆)	24300 kg _{CO2e} /kg _{SF6}

Functional unit of a car

Functional unit:

What	How much	How well	Where	How long
“Transporting	1 passenger safely		in Europe	for 1 km commuting

Key parameter: Vehicle occupancy

(1.1 passenger for commuting, 1.4 overall in DK)

→ Possible to enter the number of occupants for the car scenario
and 1 scenario electric car with 2 occupants in the sensitivity study

How to perform a carbon footprint

Amount of inputs
per functional Unit
(e.g. kg/pers-km)

Q

X

GHG Emission
factors per
unit input:

E

=

Carbon footprint
per FU

F

kg_{steel}/pers-km

x

kg_{CO2eq}/kg_{steel}

=

kg_{CO2eq}/FU

kg_{gas}/pers-km

x

kg_{CO2eq}/kg_{gas}

=

kg_{CO2eq}/FU

You determine the inputs per FU!

Database
& spreadsheet



You interpret the spreadsheet results

- Look at main contributions
- Use graph on the right
- Identify potential improvements

The 1210x QMAS Sustainability tool – Carbon footprint

Carbon footprint tool										
Functional Unit		Transporting 1 passenger safely in Denmark for 1 km commuting			Transformation multiplier		1000		t/kg, kg/g...	
FU		1 pers-km			Conversion factor		3.6		MJ/kWh	
					Conversion efficiency - Electricity		25%			
					Conversion efficiency -Heat		80%			
System 1: Gasoline car										
Data		Unit			Material		Energy content		Unit	
Lifespan of Gasoline car		150000			Incineration mix - ICEV		42.5		MJ/kg	
Weight of car		1422.1017								
Occupancy		1								
List of Processes	Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit	Quantity per FU	Unit	Carbon footprint [kg CO2eq/Unit]	Carbon footprint [kg CO2eq/FU]	fraction carbon footprint
Manufacturing (Materials + Processing)		Manufacturing (Materials + Processing)		Per vehicle		Per FU				
steel production, low-alloyed, hot rolled, RER	21143	steel production, low-alloyed, hot rolled	RER	1040	kg	6.93E-03	kg	2.27	1.57E-02	5.04%
sheet rolling, steel	20338	sheet rolling, steel	RER	780	kg	5.20E-03	kg	0.32	1.64E-03	0.53%
aluminium production, primary, ingot	502	aluminium production, primary, ingot	IAI Are	55	kg	3.67E-04	kg	7.39	2.71E-03	0.87%
Copper & other metals (Zn, Ni, Pb)	20335	sheet rolling, copper	RER	38	kg	2.53E-04	kg	0.48	1.20E-04	0.04%
platinum	14151	platinum to generic market for metal catalyst for catalytic	GLO	1.70E-03	kg	1.13E-08	kg	45758.65	5.19E-04	0.17%
polyethylene, polypropylene & PVC	17333	market for polyethylene, high density, granulate	GLO	175	kg	1.17E-03	kg	2.83	3.31E-03	1.06%
synthetic rubber	21656	market for synthetic rubber	GLO	46	kg	3.07E-04	kg	3.17	9.73E-04	0.31%
glas	9020	market for flat glass, uncoated	RoW	32	kg	2.13E-04	kg	1.06	2.27E-04	0.07%
electronics, for control units	8391	electronics production, for control units	RER	6.1	kg	4.07E-05	kg	31.76	1.29E-03	0.41%
ethylene, paints, fluids	8639	market for ethylene	RER	30	kg	2.00E-04	kg	1.82	3.65E-04	0.12%
tap water	21711	tap water production, conventional treatment	RoW	3400	kg	2.27E-02	kg	0.00	9.42E-06	0.00%
heat, district or industrial, natural gas	10352	heat production, natural gas, at industrial furnace >100kW	Europe	2400	MJ	1.60E-02	MJ	0.08	1.21E-03	0.39%
electricity, medium voltage	7666	market group for electricity, low voltage	RER	2250	kWh	1.50E-02	kWh	0.33	4.96E-03	1.59%
Transport		Transport		Per vehicle		Per FU				
transport, freight train	22465	market for transport, freight, train, fleet average	Europe	371	km	3.52E-03	metric ton*km	0.05	1.64E-04	0.00
transport, freight, lorry 16-32 metric ton	22337	market for transport, freight, lorry, 16-32 metric ton, diesel	RER	39	km	3.70E-04	metric ton*km	0.19	7.01E-05	0.00
transport, freight, sea, container ship	22445	market for transport, freight, sea, container ship, heavy fuel oil	GLO	8000	km	7.58E-02	metric ton*km	0.01	7.78E-04	0.00
Use phase		Use phase		Per V-km		Per FU				
Supply chain petrol, low-sulfur	16715	market for petrol, low-sulfur	Europe	0.0619	kg	6.19E-02	kg	1.04	6.42E-02	0.21
Direct combustion	100001	Emissions & operation of vehicle	RER	0.0619	kg petrol	6.19E-02	kg petrol	3.43	2.12E-01	0.68
Disposal		Disposal		Per vehicle		Per FU				
plastic mixture incineration	23993	treatment of waste plastic, mixture, municipal incineration	CH	221.00	kg	1.47E-03	kg	2.37	3.49E-03	0.01
Glass to landfill	23563	treatment of waste glass, inert material landfill	CH	283.00	kg	1.89E-03	kg	0.01	1.05E-05	0.00
Avoided electricity plastic incineration	7666	market group for electricity, low voltage	RER	221.00	kWh	-4.35E-03	kWh	0.33	-1.44E-03	0.00
								Total kgCO2eq/FU	0.312	

Case study: Gasoline vs electric vehicle vs bike or e-bike



The ecoinvent database 25000+ processes – in the ecoinvent tab

COPYRIGHTS: The present Ecoinvent data can only be used within DTU courses or theses. It cannot					Method	IMPACT World+ v2.1, footprint version
Identify the # number		Regions: RER, UCTE: Europe ROW: Rest of the World GLO: Global CA: Canada IN: India			Category	climate change
→ Copy in selected Processes & in Carbon footprint		CN: China US: USA	Olivier Jean Joliet: You might use the arrow on the bottom right of this field to then enter keywords and search for processes of interest		Indicator	carbon footprint
Process # ↓	Activity Name	Geography	Reference Product Name	Refer ence Prod	Reference Product Amount	kg CO2-Eq
6697	market group for electricity, high voltage	BR	electricity, high voltage	kWh	1	0.13
6698	market group for electricity, high voltage	CA	electricity, high voltage	kWh	1	0.18
6699	market group for electricity, high voltage	CN	electricity, high voltage	kWh	1	0.84
6700	market group for electricity, high voltage	CN-SGCC	electricity, high voltage	kWh	1	0.89
6701	market group for electricity, high voltage	Canada without Quebec	electricity, high voltage	kWh	1	0.27
6702	market group for electricity, high voltage	ENTSO-E	electricity, high voltage	kWh	1	0.32
6703	market group for electricity, high voltage	Europe without Switzerland	electricity, high voltage	kWh	1	0.33
6704	market group for electricity, high voltage	GLO	electricity, high voltage	kWh	1	0.67
6705	market group for electricity, high voltage	IN	electricity, high voltage	kWh	1	1.31
6706	market group for electricity, high voltage	RAF	electricity, high voltage	kWh	1	0.75
6707	market group for electricity, high voltage	RAS	electricity, high voltage	kWh	1	0.85
6708	market group for electricity, high voltage	RER	electricity, high voltage	kWh	1	0.33
6709	market group for electricity, high voltage	RLA	electricity, high voltage	kWh	1	0.31
6710	market group for electricity, high voltage	RME	electricity, high voltage	kWh	1	0.78
6711	market group for electricity, high voltage	RNA	electricity, high voltage	kWh	1	0.44
6712	market group for electricity, high voltage	UCTE	electricity, high voltage	kWh	1	0.36
6713	market group for electricity, high voltage	US	electricity, high voltage	kWh	1	0.48

How to perform a carbon footprint – 12100 spreadsheet

Collect the
product input data
You!

Column E

e.g. $1040 \text{ kg}_{\text{steel}}/\text{V}$ $\times 1/150,000\text{km}$
 $6.2 \text{ kg}/100 \text{ V-km}$ $\times 1$

Transform them **per**
functional unit
FU= 1 pers-km **You**

Column G

$0.0069 \text{ kg}_{\text{steel}}/\text{pers-km}$ $\times$
 $0.062 \text{ kg}_{\text{gas}}/\text{pers--km}$ $\times$

Find **GHG emission**
factors per unit input:
in database

Column B → I

$2.2 \text{ kg}_{\text{CO2eq}}/\text{kg}_{\text{steel}}$
 $4.45 \text{ kg}_{\text{CO2eq}}/\text{kg}_{\text{gas}}$

Multiply **F x H**
to get the **carbon**
footprint per FU

Column J

$= 0.015 \text{ kg}_{\text{CO2eq}}/\text{FU}$
 $= 0.275 \text{ kg}_{\text{CO2eq}}/\text{FU}$

Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit
	Manufacturing (Materials + Processing)		Per vehicle	
21143	steel production, low-alloyed, hot rolled	RER	1040	kg
20338	sheet rolling, steel	RER	780	kg
502	aluminium production, primary, ingot	IAI Are	55	kg
20335	sheet rolling, copper	RER	38	kg
14151	platinum to generic market for metal catalyst for catalytic	GLO	1.70E-03	kg
17333	market for polyethylene, high density, granulate	GLO	175	kg
21656	market for synthetic rubber	GLO	46	kg
9020	market for flat glass, uncoated	RoW	32	kg
8391	electronics production, for control units	RER	6.1	kg
8639	market for ethylene	RER	30	kg
21711	tap water production, conventional treatment	RoW	3400	kg
10352	heat production, natural gas, at industrial furnace >100kW	Europe	2400	MJ
7666	market group for electricity, low voltage	RER	2250	kWh
	Transport		Per vehicle	
22465	market for transport, freight, train, fleet average	Europe	371	km
22337	market for transport, freight, lorry, 16-32 metric ton, diesel	RER	39	km
22445	market for transport, freight, sea, container ship, heavy freight	GLO	8000	km
	Use phase		Per V-km	
16715	market for petrol, low-sulfur	Europe	0.0619	kg
100001	Emissions & operation of vehicle	RER	0.0619	kg petrol
	Disposal		Per vehicle	
23993	treatment of waste plastic, mixture, municipal incineration	CH	221.00	kg
23563	treatment of waste glass, inert material landfill	CH	283.00	kg
7666	market group for electricity, low voltage	RER	221.00	kWh

Quantity per FU	Unit
Per FU	
6.93E-03	kg
5.20E-03	kg
3.67E-04	kg
2.53E-04	kg
1.13E-08	kg
1.17E-03	kg
3.07E-04	kg
2.13E-04	kg
4.07E-05	kg
2.00E-04	kg
2.27E-02	kg
1.60E-02	MJ
1.50E-02	kWh
Per FU	
3.52E-03	metric ton*km
3.70E-04	metric ton*km
7.58E-02	metric ton*km
Per FU	
6.19E-02	kg
6.19E-02	kg petrol
Per FU	
1.47E-03	kg
1.89E-03	kg
-4.35E-03	kWh

Process ID
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21656
9020
8391
8639
21711
10352
7666
22465
22337
22445
16715
100001
23993
23563
7666



Carbon footprint [kg CO2eq/Unit]
2.27
0.32
7.39
0.48
45758.65
2.83
3.17
1.06
31.76
1.82
0.00
0.08
0.33
0.05
0.19
0.01
1.04
3.43
2.37
0.01
0.33

Carbon footprint [kg CO2eq/FU]
1.57E-02
1.64E-03
2.71E-03
1.20E-04
5.19E-04
3.31E-03
9.73E-04
2.27E-04
1.29E-03
3.65E-04
9.42E-06
1.21E-03
4.96E-03
1.64E-04
7.01E-05
7.78E-04
6.42E-02
2.12E-01
3.49E-03
1.05E-05
-1.44E-03
0.312

Sum up →

The ecoinvent database 25000+ processes – in the ecoinvent tab

COPYRIGHTS: The present Ecoinvent data can only be used within DTU courses or theses. It cannot					Method	IMPACT World+ v2.1, footprint version
Identify the # number		Regions: RER, UCTE: Europe ROW: Rest of the World GLO: Global CA: Canada IN: India			Category	climate change
→ Copy in selected Processes & in Carbon footprint		CN: China US: USA	Olivier Jean Joliet: You might use the arrow on the bottom right of this field to then enter keywords and search for processes of interest		Indicator	carbon footprint
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6701	market group for electricity, high voltage	Canada without Quebec	electricity, high voltage	kWh	1	0.27
6702	market group for electricity, high voltage	ENTSO-E	electricity, high voltage	kWh	1	0.32
6703	market group for electricity, high voltage	Europe without Switzerland	electricity, high voltage	kWh	1	0.33
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6712	market group for electricity, high voltage	UCTE	electricity, high voltage	kWh	1	0.36
6713	market group for electricity, high voltage	US	electricity, high voltage	kWh	1	0.48



Scenario comparison and interpretation

Interpretation Principles

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Interpretation question for each scenario

Which are the largest contribution to the environmental impact in terms of production phase, in terms of pollutant?

- The grouping of processes can be arbitrary and depends on how the system is modeled.
- Also consider large sets of individual processes that each appear to have small impacts but lead to substantial impacts when summed

Which life cycle stage has the highest potential to reduce impacts (also depending on feasibility)

- In some win-win cases, both impacts and costs can be reduced at the same time.
- Even a limited low-cost intervention can be extremely efficient in reducing impacts.
- **If some results are surprising:** WHY? What are the underlying parameter components and stages responsible for it?
*Never let it go until you understand the rational for a difference or a surprising results.
Either you made a mistake ... or you will learn something!*



Download the car solutions



12100 Quantitative methods to assess sustainabil...



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▼ W4: Carbon footprint and environmental indicators



Module description

▼ Studies before

▶ Video lectures



▼ Exercise



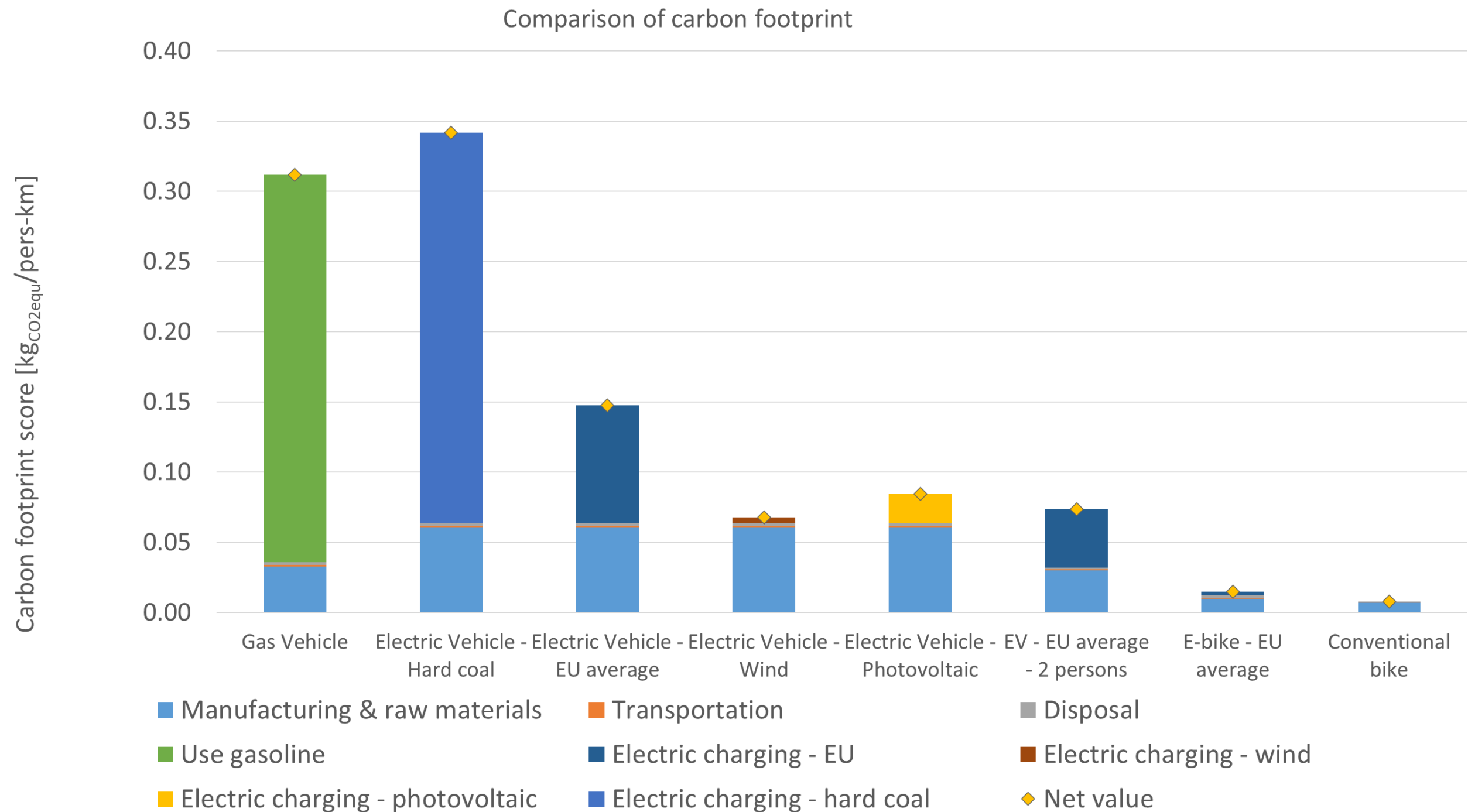
Exercise - Carbon footprint, Mobility



Exercise solutions - Carbon footprint,...

Exercise

Carbon footprint: gasoline vs electric vehicle



Interpret the results of the car example

1. What are the three main processes dominating the impacts in the different scenarios?

Carbon footprint [kg CO2eq/FU]	fraction carbon footprint
1.55E-02	4.96%
1.69E-03	0.54%
2.75E-03	0.88%
1.18E-04	0.04%
5.17E-04	0.17%
3.27E-03	1.05%
9.65E-04	0.31%
2.30E-04	0.07%
1.30E-03	0.42%
3.52E-04	0.11%
9.55E-06	0.00%
1.23E-03	0.39%
4.82E-03	1.55%
1.62E-04	0.00
7.09E-05	0.00
7.83E-04	0.00
0.00E+00	0.00
6.37E-02	0.20
2.12E-01	0.68
3.55E-03	0.01
9.99E-06	0.00
-1.40E-03	0.00
0.312	

2. How to further reduce the impacts of an electric car?

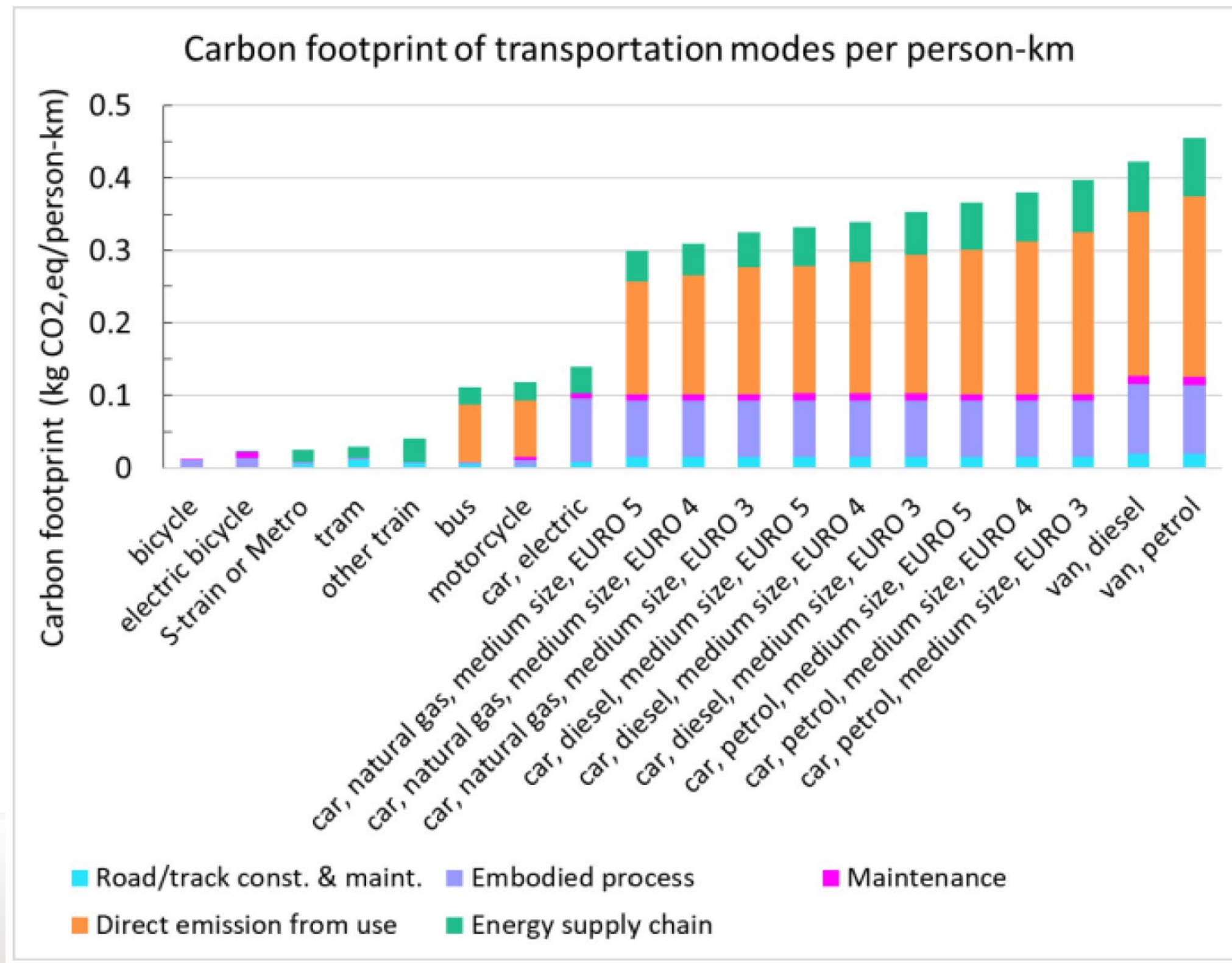


3. What can individuals do in priority to reduce the carbon footprint of person transportation?

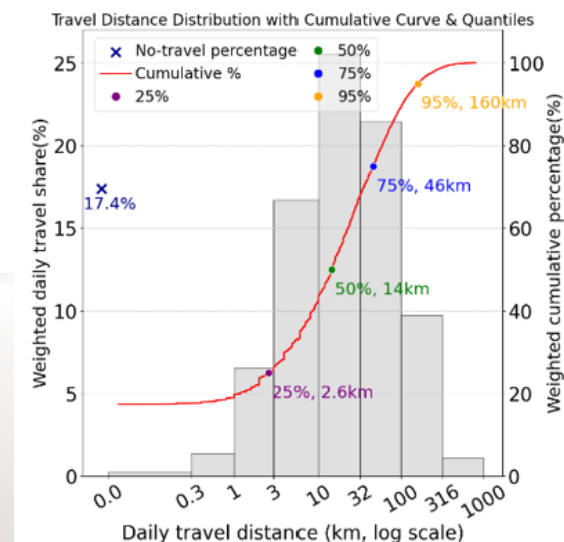
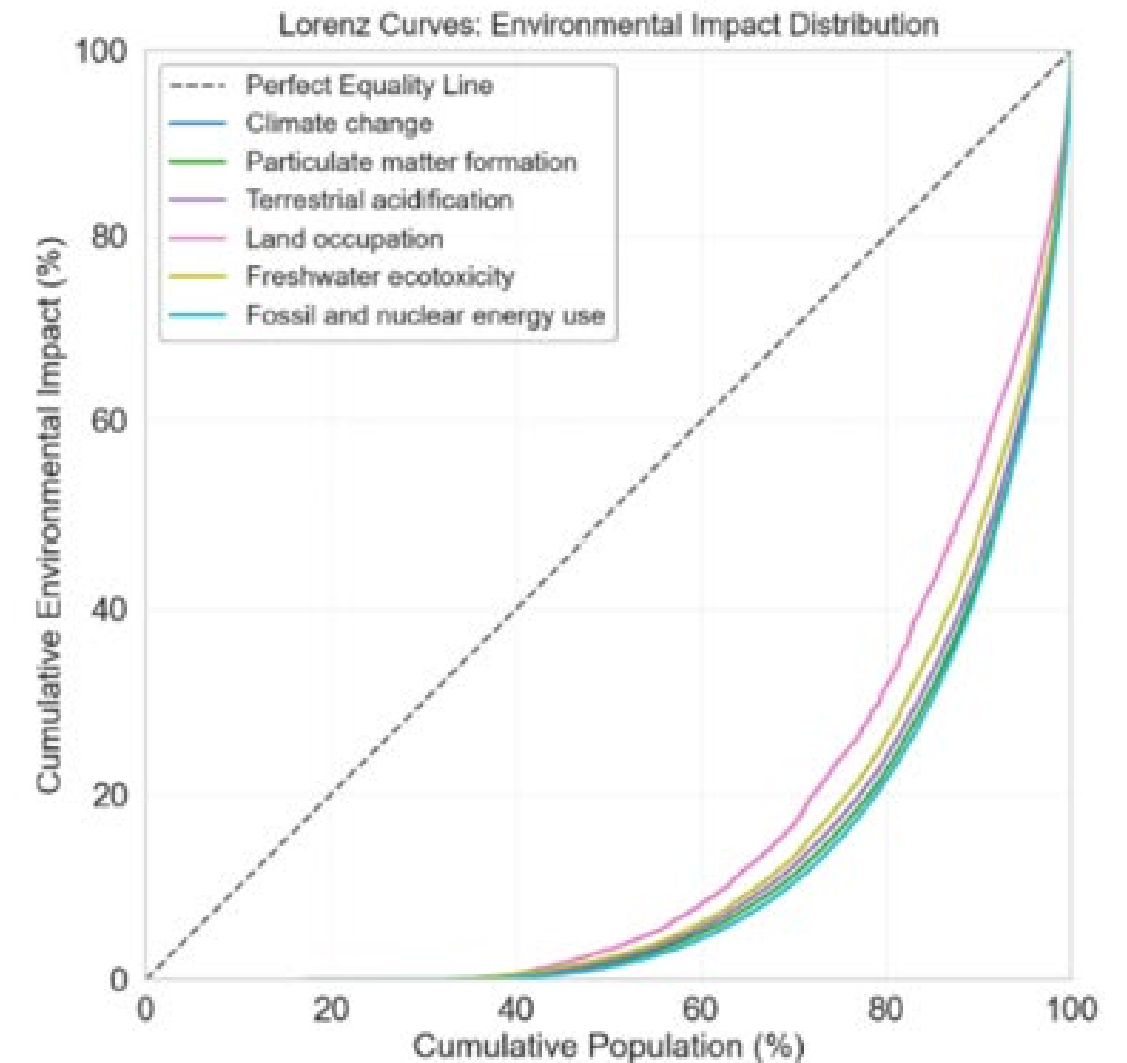
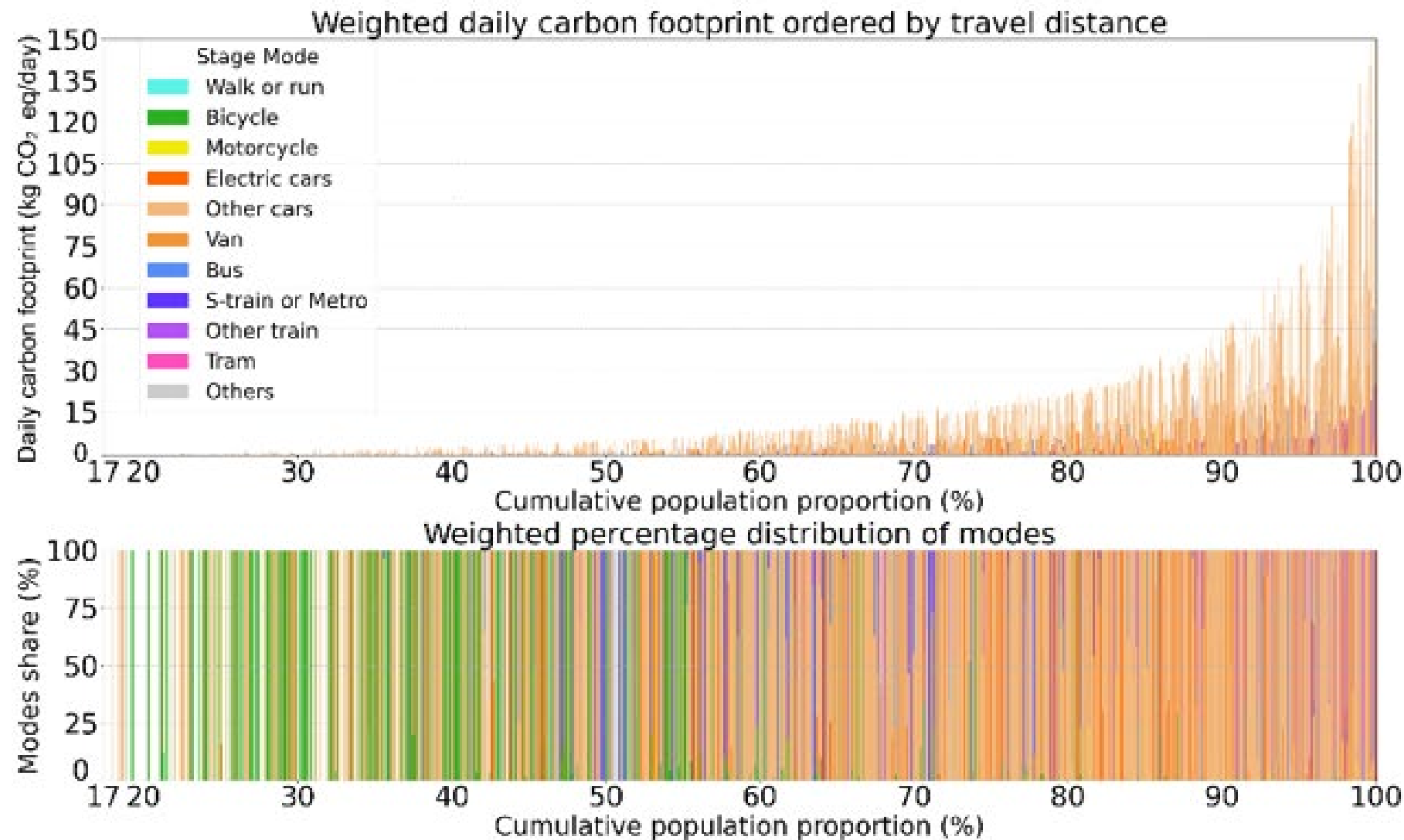


- Steel material and manufacturing
- Platinum
- Electronics
- HDPE- high density polyethylene
- Direct combustion
- Supply chain petrol / gasoline
- Plastic incineration
- Manufacturing electricity
- Ship transportation from Asia
- Paints & ethylene

Per pers-km carbon footprint for different modes of transport



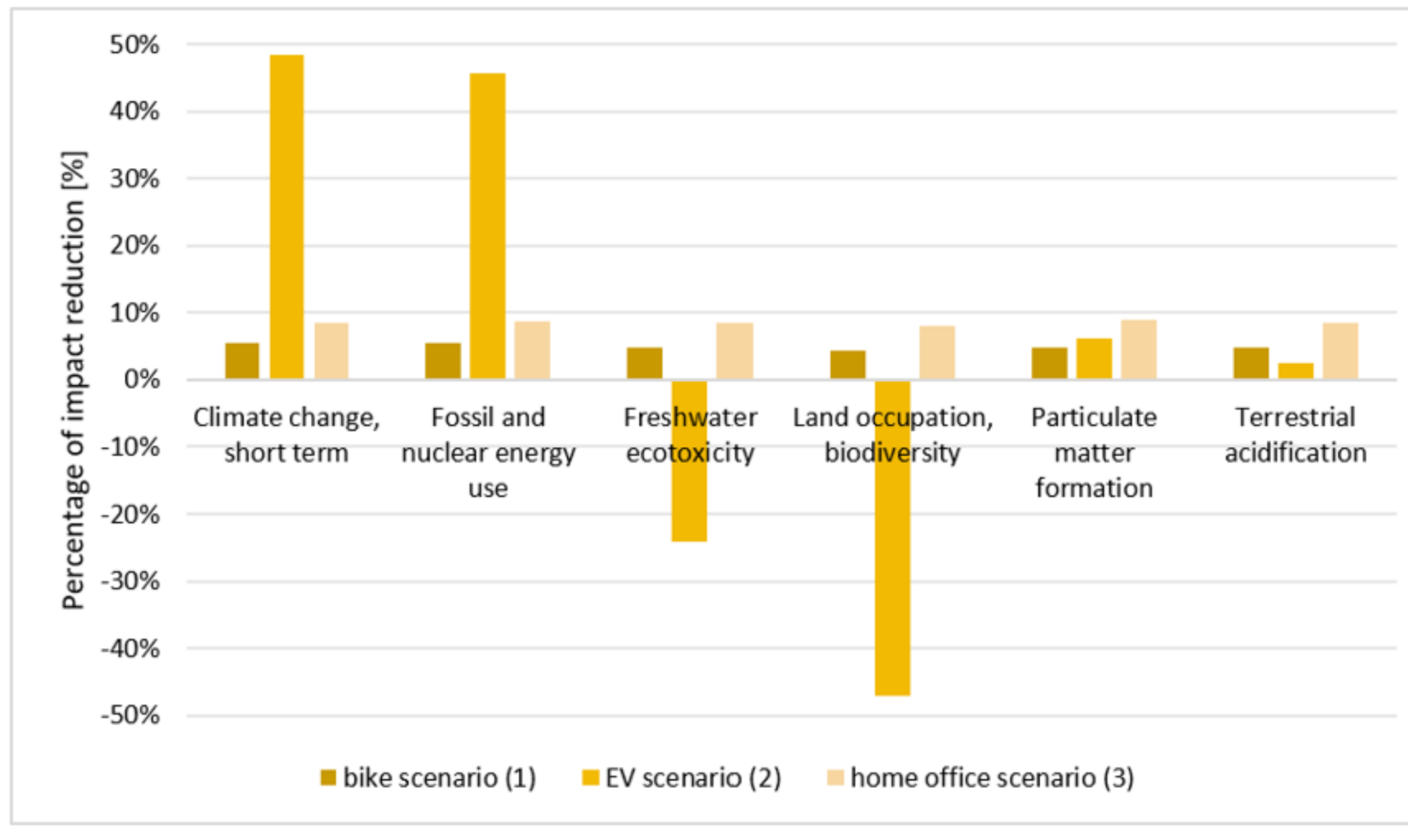
Daily carbon footprint – Danish population



Weinan Yan & Marion George, 2026

Alternative scenarios for impact reduction

Three alternative scenarios are modelled and tested in this section, 1) all short-distance car trips (less than 10km) made by bike (bike scenario), 2) all cars and vans being electric (EV scenario) and 3) one more day of home office when possible (home office scenario).





Carbon footprint practice

Morning coffee – important for case study and exam

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Learning Objectives

- ▶ Define a process to establish a carbon footprint
- ▶ Compare CO₂ for various materials and processes
- ▶ Perform the carbon footprint of a morning coffee
- ▶ Interpret results, comparing scenario performances and identifying hot spots

Defining the basis for comparison / functional unit based on 5 questions

**HOW MUCH/
HOW MANY?**

**WHICH FUNCTION
& HOW WELL?**

**FOR
WHAT?**

FOR HOW LONG

(When needed)

Positioning
property

WHERE?

Functional
unit (FU):

xxx



What are the processes in a cup of coffee life cycle – design the process tree

Case study: Morning coffee

In group, select the three main processes that dominate the carbon footprint of a cup of coffee ?



Coffee bean production

Roasting

Transport

Packaging

Brewing

Cups

Water

Electricity for warming water

Go on module 4, teaching session, Exercise – Carbon footprint, Coffee



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▼ Teaching session

Exercise - Carbon footprint, Coffee

✓

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...





Teaching session

Activity 1: Carbon footprint of a morning coffee



- ▶ What are the inputs contributing to the carbon footprint of a cup of roasted coffee?
- ▶ To have a cup of coffee (1 cup is 1 dl = 0.1 liters), you need 11.5 grams of coffee green beans.
- ▶ Coffee needs to be roasted which requires 2.31 MJ of heat from natural gas and 0.11 kWh DK electricity per kg coffee.
- ▶ Hot water heating requires 0.12 kWh per liter water and 0.028 kWh per cup for the coffee machine stand-by.
- ▶ Coffee is transported over 3500 km by truck and 4500 km by sea freight. At its end of life, paper packaging and the coffee maker are incinerated with energy recovery.
- ▶ Download and open file "file "Exercise - Carbon footprint, Coffee 1210X QMAS" filling in the green/turquoise cells with missing data in rows 21 to 43

Perform the carbon footprint – excel tool

1. Search for the process number in the ecoinvent tab.

Enter these in column A of the “selected processes” and “carbon footprint” tabs

Process	Process #	Environment
coffee green bean production, robusta, RoW		Materials
coffee production, arabica, BR-MG		
market for coffee, green bean	2073	coffee, green bean
		coffee maker
		tion, natural gas, at b
		aluminium, primary, in
		raft paper
		processes
		electricity, low voltage

List of Processes	Process ID
Manufacturing (Materials + Processing)	
Coffee green beans production robusta (11.5g)	
Heat from natural gas for coffee roasting	10063
Electricity for coffee roasting	
Coffee machine (1 unit of 3 kg)	3057
Paper Cup (8g paper)	12216
Packaging (capsule 2g alu - aluminium)	523
Transportation	
Coffee, transport, freight, lorry 16-32 metric ton (3500 km)	22338
Coffee transport, freight, sea, container ship, (4500 km)	22446
Use phase	
Tap water (0.1 L/cup)	21695
Electricity hot water (0.12kWh/L)	0
Stand-by electricity (0.028 kWh/cup)	0
Disposal	
Aluminium sent to sanitary landfill	23363
Coffee maker as plastic mixture incineration	23957
Paper cup sent to incineration	23743
Avoided heat paper and plastic incineration	10063
Avoided electricity paper and plastic incineration	0

2. Enter in column G the amount of units of each process per FU, i.e per 1 cup of coffee

Check your units!

3. Multiply $G \times I$ to get the carbon footprint per FU in J

4. Interpret

What are the dominant processes?

Rank from
highest to
lowest

G		x		I		=		J	
Unit	Quantity per FU			Carbon footprint per unit [kg CO ₂ eq/Unit]				Carbon footprint per FU [kg CO ₂ eq/FU]	
	per FU								
kg/serving				#N/A					
MJ/kg coffee	0.00E+00			0.07				0.00E+00	
kWh/kg coffee				#N/A				#N/A	
unit				41.08				0.00E+00	
kg/serving	8.00E-03			0.55				4.38E-03	
kg/serving	2.00E-03			10.18				2.04E-02	
	per FU								
km	0.00E+00			0.20				0.00E+00	
km	0.00E+00			0.01				0.00E+00	
	per FU								
kg/serving	1.00E-01			0.00				3.02E-05	
kWh/liter				#N/A				#N/A	
kWh/serving	2.80E-02			#N/A				#N/A	
maker	per FU								
kg/serving	2.00E-03			0.02				3.43E-05	
kg/coffee maker	#DIV/0!			2.34				#DIV/0!	
kg/serving				0.01				0.00E+00	
	#DIV/0!			0.07				#DIV/0!	
	#DIV/0!			#N/A				#N/A	
				Total kgCO ₂ eq/FU				#N/A	

← Sum up

A. Process identification: Answer the following questions

1. Identify the processes for coffee bean production in the ecoinvent Tab, then enter then in the selected process and carbon footprint tabs

a) What is this process number and "coffee green bean production, robusta" produced in the Rest of the World (RoW).

b) What is the corresponding carbon footprint for 1 kg of this type of coffee, in $\text{kg}_{\text{CO}_2\text{eq}}/\text{kg}_{\text{coffee}}$?

2. Be curious and also look at different types of coffee (search for coffee in the ecoinvent tab).

a) What is the range in carbon footprint values from lowest to highest carbon footprint of different types and provenance of coffee grains.

b) Is ROW robusta beans at the low or the high end of the range?

B. Amounts per FU: Answer the following questions:

3. Fill the amount of each process per cup of coffee (per FU) in column G

- a) *What is the amount of coffee per FU?*
- b) *What is the amount of electricity per FU for coffee roasting?*
- c) *What is the amount of electricity per FU to heat the water?*
- d) *How many coffee machines you need per FU?*
- e) *What is the process for DK electricity and its carbon footprint per kWh?*

C. Carbon Footprint of Robusta: Answer the following questions

4. Look at the carbon footprint of the robusta coffee

a) *What is the total carbon footprint contribution of the bean production in kgCO₂e/cup of coffee?*

b) *What is the total carbon footprint of this cup of coffee in kgCO₂ per kg?*

c) *What are the 3 most contributing processes to this carbon footprint ... how is this different from your first assessment?*

- Coffee, transport by container ship
- Coffee, transport by truck/lorry
- Electricity for hot water & stand by
- Coffee green bean production (robusta RoW)
- Heat from natural gas for coffee roasting
- Coffee machine (1 unit of 3 kg)
- Paper cup
- Aluminium packaging

D. Comparison of different coffee scenarios and improvements

First complement the electricity mix for roasting and heating the water in each of the three scenarios, then answer the question

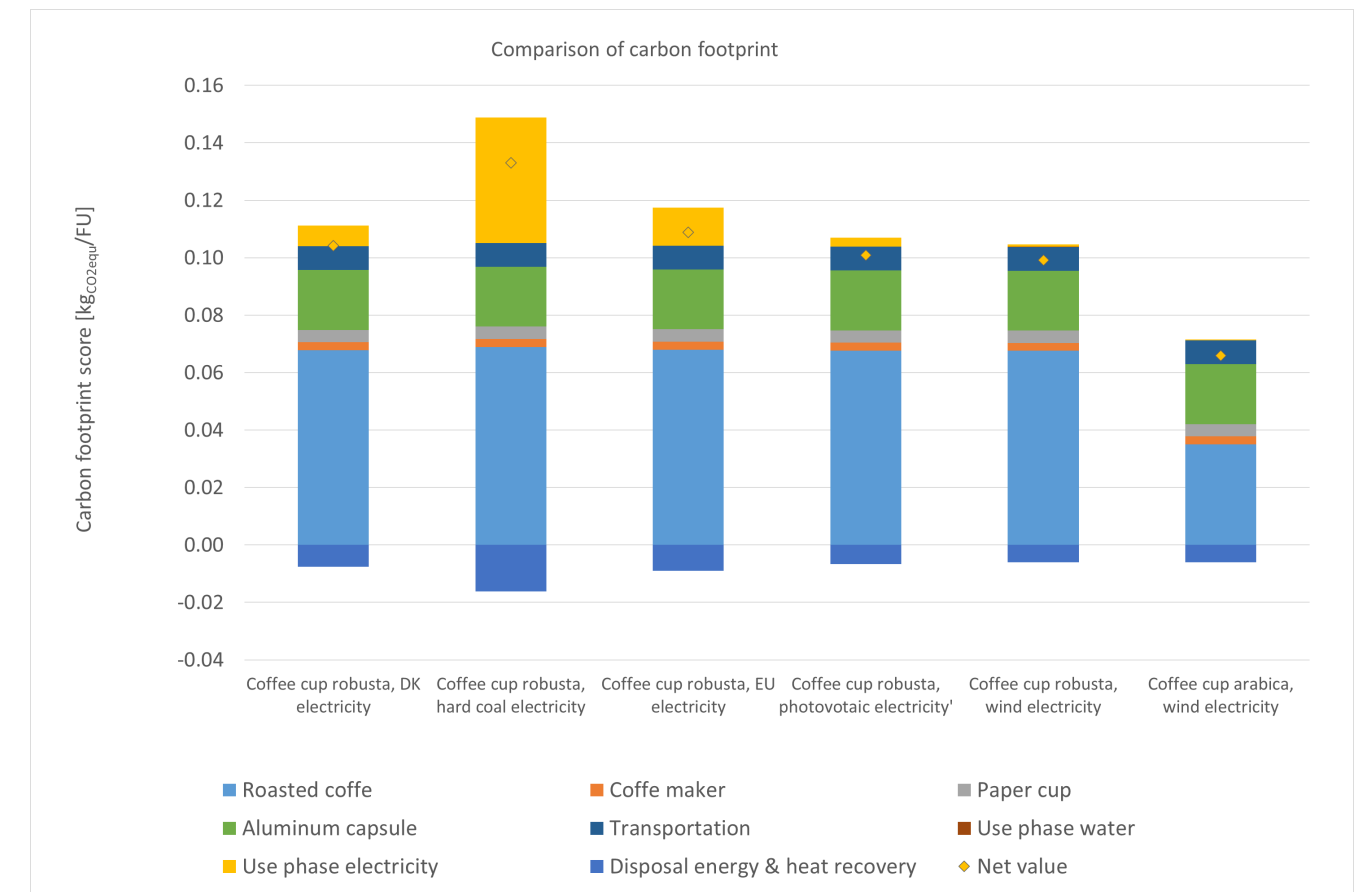
5. Compare the different types of coffee for the considered scenarios on the carbon footprint graph. What do you learn?

6. Propose three measures that could reduce the carbon footprint of a cup of coffee

a) Xxx

b) Xxx

c) Xxx



→ You are ready to perform the self-assessment on your own! After answering its last question, you will also obtain the solution for the spreadsheet file coffee exercise